FD Calculator Service - Code Audit Report

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Service: FD Calculator Microservice
Port: 8084
API Prefix: /api/v1/fd
Technology Stack: Spring Boot 3.3.4, Java 21, Maven, MySQL, Kafka, Spring Security OAuth2

Executive Summary

The FD Calculator Service is a simulation-only Fixed Deposit maturity estimation microservice. It calculates and explains estimated FD maturity values without creating accounts, moving funds, or scheduling financial events. The service has been standardized with the Product & Pricing Service for consistency in configuration, observability, and deployment.

Architecture Overview

Service Communication

Angular UI !' API Gateway !' FD Calculator (8084)
!“
Product Service (8080)
Customer Service (8082)

Key Integrations

- **Product Service:** Fetches product configuration and rates via REST
- **Customer Service:** Fetches customer categories via REST
- **Kafka:** Batch calculation logging and audit events
- **MySQL:** Calculation audit persistence

API Endpoints

Public Endpoints

Configuration

Environment Variables

- `DB_URL` - Database connection URL
- `DB_USER` - Database username

- `DB_PASSWORD` - Database password
- `JWT_SECRET` - JWT secret key (dev only)
- `JWT_ISSUER_URI` - JWT issuer URI (default: http://localhost:8081)
- `PRODUCT_SERVICE_URL` - Product Service URL (default: http://localhost:8083)
- `CUSTOMER_SERVICE_URL` - Customer Service URL (default: http://localhost:8082)
- `KAFKA_BOOTSTRAP_SERVERS` - Kafka bootstrap servers (default: localhost:9092)
- `SERVER_PORT` - Server port (default: 8084)

Standardized Configuration

- Profile-based configuration (dev, prod)
- JPA ddl-auto: validate (prod), update (dev)
- Actuator endpoints: health, info, metrics, prometheus
- Structured JSON logging with correlation IDs
- Cache configuration for product rates
- Category bonus configuration

Security

Authentication

- OAuth2 Resource Server
- JWT validation
- HMAC-256 (dev) or JWKS (prod)

Authorization

- Role-based access control
- CUSTOMER role required for all endpoints
- Method-level security with @PreAuthorize

CORS

- Allowed origins: http://localhost:4200, http://127.0.0.1:4200
- To be removed when API Gateway is implemented

Observability

Structured Logging

- JSON structured logging via Logstash Encoder (logstash-logback-encoder 7.4)
- Correlation IDs in all logs
- Log files written to ./logs/fd-calculator.log
- 30-day log retention
- Configuration: logback-spring.xml

Metrics

- Actuator endpoints: /actuator/health, /actuator/metrics, /actuator/prometheus
- Custom business metrics (MetricsConfig):

- fd.calculations.total - Total calculations
- fd.calculations.errors - Calculation errors
- fd.calculations.duration - Calculation duration timer
- fd.cache.hits - Cache hits
- fd.cache.misses - Cache misses

Kafka Integration

- Topics:

- fd-calculator-batch-calculations - Batch calculation results
- fd-calculator-audit-logs - Audit log events

- Environment variable: KAFKA_BOOTSTRAP_SERVERS (default: localhost:9092)

Data Model

CalculationAudit Entity

- Table: fd_calculation_audit
- Primary Key: id (String with CALC- prefix)
- Persistent UUID: uuid (String)
- Customer: customerId (String)
- Product: productCode (String)
- Financial: principal, termMonths, baseRate, effectiveRate, interestEarned, maturityAmount
- Audit Columns: createdAt, updatedAt, uuid, oooUserId, oooWsId, ldbid, oooAcptTs, oooHostTs, oooLocalTs, oooAcptTsUteOfst, ruleSystemId
- Uses @PrePersist and @PreUpdate hooks for automatic field population

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Conventions

Naming Conventions

- Java fields: camelCase
- Database columns: snake_case
- API endpoints: kebab-case
- DTOs: Java Records for immutability
- Services: Interface + Impl pattern
- Exceptions: Descriptive names ending with "Exception"

Code Conventions

- Use Java Records for DTOs (immutable)

- Use Lombok for entities (builder pattern)
- Use @Valid for request validation
- Use @NotNull, @NotBlank, @DecimalMin, @Digits for field validation
- Use @Operation and @ApiResponse for Swagger documentation
- Use @SecurityRequirement for endpoint security

Monetary Values

- All monetary values use BigDecimal
- Currency amounts: scale 2
- Rates: scale 2-6
- Rounding mode: HALF_UP

Key Components

Controllers

- FdCalculationController - REST API endpoints

Services

- FdCalculatorService - Main business logic
- CalculationEngine - Core FD calculation logic
- ProductCatalogService - Product rate caching
- CategoryBonusService - Category-based rate bonuses
- ExplainService - Plain-language explanations
- ExportService - CSV export
- AuditService - Calculation persistence

Repositories

- CalculationAuditRepository - JPA repository for audit trail

DTOs (Java Records)

- CalculationRequest, ComparisonRequest, TermDto
- CalculationResponse, ComparisonResponse, ProductPreviewResponse
- BatchCalculationRequest, BatchCalculationResponse

Enums

- PayoutMode: CUMULATIVE, PERIODIC
- InterestType: SIMPLE, COMPOUND
- CompoundingFrequency: DAILY, MONTHLY, QUARTERLY, YEARLY
- TermUnit: DAYS, MONTHS, YEARS
- CustomerCategory: GENERAL, SENIOR_CITIZEN, EMPLOYEE, CORPORATE

Dependencies

Core Spring

- Spring Boot Starter Web
- Spring Boot Starter Validation
- Spring Boot Starter Security
- Spring Boot Starter OAuth2 Resource Server
- Spring Boot Starter Actuator

Database

- Spring Boot Starter Data JPA
- MySQL Connector
- H2 Database (dev/testing)

External Services

- Spring Cloud OpenFeign
- Spring Boot Starter Kafka

Documentation

- Springdoc OpenAPI Starter WebMVC UI

Utilities

- Lombok
- MapStruct
- JJWT

Observability

- Micrometer Core
- Logstash Logback Encoder 7.4

Deployment

Docker

- Base image: eclipse-temurin:21-jre-alpine
- Non-root user: appuser
- Container support enabled
- MaxRAMPercentage: 75%
- Port: 8084

Build

mvn clean package

Run

docker build -t fd-calculator .
docker run -p 8084:8084 fd-calculator

Testing

Test Coverage

- CalculationEngineImplTest - Simple, compound, rounding, edge values
- CategoryBonusServiceImplTest - Category bonuses, caps
- FdCalculatorServiceImplTest - Happy paths, inactive, not found, invalid term
- FdCalculationControllerTest - 401, 403, 400 validation, 200 response
- FdCalculatorIntegrationTest - Full context + WireMock + H2

Running Tests

mvn test
mvn verify

Integration Points

Product Service

- Endpoint: /internal/products/{productCode}/calculator-config
- Purpose: Fetch product configuration
- Cached: Yes (Caffeine)
- Fallback: Mock client (dev)

Customer Service

- Endpoint: /customers/{customerId}/category
- Purpose: Fetch customer category
- Cached: No
- Fallback: Mock client (dev)

Kafka

- Topics: fd-calculator-batch-calculations, fd-calculator-audit-logs
- Purpose: Batch processing and logging

Recent Updates

Standardization Improvements

1. **Actuator Configuration** - Added Prometheus metrics export, standardized endpoint exposure
2. **Structured Logging** - Added JSON logging with Logstash Encoder, correlation IDs, 30-day retention
3. **Metrics** - Added custom business metrics via MetricsConfig
4. **Audit Entity** - Standardized @PrePersist/@PreUpdate hooks for audit columns
5. **Docker** - Created Dockerfile matching Product Service structure
6. **API Documentation** - Standardized security scheme name to bearerAuth

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Compliance

Banking Standards

- Audit trail with 11 standardized columns
- UUID for persistent identification
- Ownership enforcement
- Calculation ID for business key
- Timestamp tracking (acceptance, host, local)

Security Standards

- JWT-based authentication
- Role-based authorization
- Stateless session management
- CORS configuration (dev only)

Future Enhancements

- WebSocket for real-time calculation updates
- Additional export formats (PDF, Excel)
- Advanced comparison features
- Historical calculation trends
- Calculation templates
- Multi-currency support

Documentation

- Swagger UI: http://localhost:8084/swagger-ui/index.html
- OpenAPI Spec: http://localhost:8084/v3/api-docs
- Health: http://localhost:8084/actuator/health
- Metrics: http://localhost:8084/actuator/metrics

- Prometheus: http://localhost:8084/actuator/prometheus

Contact

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End of Code Audit Report